

Instructor Notes:



Web basics-JavaScript

Lesson 2: JavaScript Language

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Instructor Notes:

Lesson Objectives

- Data Types and Variables
- JavaScript Operators
- Control Structures and Loops
- JavaScript Functions



Instructor Notes:

2.1: Data Types and Variables

Data Types in JavaScript

- **JavaScript is a free-form language. You do not have to declare all variables, classes, and methods**
- **Variables in JavaScript can be of type:**
 - Number (4.156, 39)
 - String ("This is JavaScript")
 - Boolean (true or false)
 - Null (null)

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Data Types in JavaScript:

Although the number of data types is small, they are sufficient for the tasks that JavaScript performs. Notice that there is no distinction between integers and real numbers; both types are just numbers. JavaScript does not provide an explicit data type for a date. However, there are related functions and a built-in date object that enable the Web page designer to manage dates.

Instructor Notes:

2.1: Data Types and Variables

Data Types in JavaScript (Contd..)

- **JavaScript variables are said to be loosely typed**
- **Defining variables:** `var variableName = value`
- **JavaScript variables**
 - Can include letters of the alphabet, digits 0-9 and the underscore (`_`) character and is case-sensitive.
 - Cannot include spaces or any other punctuation characters.
 - First character of the variable name must be either a letter or the underscore character.
 - No official limit on the length of a variable name, but must fit within a line.

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**Defining Variables:**

There are specific rules you must follow when choosing variable names.

Variable names can include letters of the alphabet, both upper and lowercase.

They can also include the digits 0-9 and the underscore (`_`) character.

Variable names cannot include spaces or any other punctuation characters.

The first character of the variable name must be either a letter or the underscore character.

Variable names are case-sensitive; `totalnum`, `Totalnum`, and `TotalNum` are separate variable names.

There is no official limit on the length of a variable name, but it must fit within one line.

JavaScript variables are said to be loosely typed. To declare a variable for a JavaScript program, you would write this:

```
var variablename = value ;
```

2.2: JavaScript Operators

Arithmetic Operator

Operator	Description	Example	Result
+	Addition	2 + 2	4
-	Subtraction	5 - 2	3
*	Multiplication	4 * 5	20
/	Division	5 / 2	2.5
%	Modulus	10 % 8	2
++	Increment	x = 5; x++	x = 6
--	Decrement	x = 5; x--	x = 4

2.2: JavaScript Operators

Comparison Operator

Operator	Description	Example	Result
==	is equal to	5 == 8	false
!=	is not equal	5 != 8	true
>	is greater than	5 > 8	false
<	is less than	5 < 8	true
>=	is greater or equal	5 >= 8	false
<=	is less or equal	5 <= 8	true

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2.2: JavaScript Operators

Assignment Operator

Operator	Example	Is same as
<code>+=</code>	<code>x += y</code>	<code>x = x + y</code>
<code>-=</code>	<code>x -= y</code>	<code>x = x - y</code>
<code>*=</code>	<code>x *= y</code>	<code>x = x * y</code>
<code>/=</code>	<code>x /= y</code>	<code>x = x / y</code>
<code>%=</code>	<code>x %= y</code>	<code>x = x % y</code>

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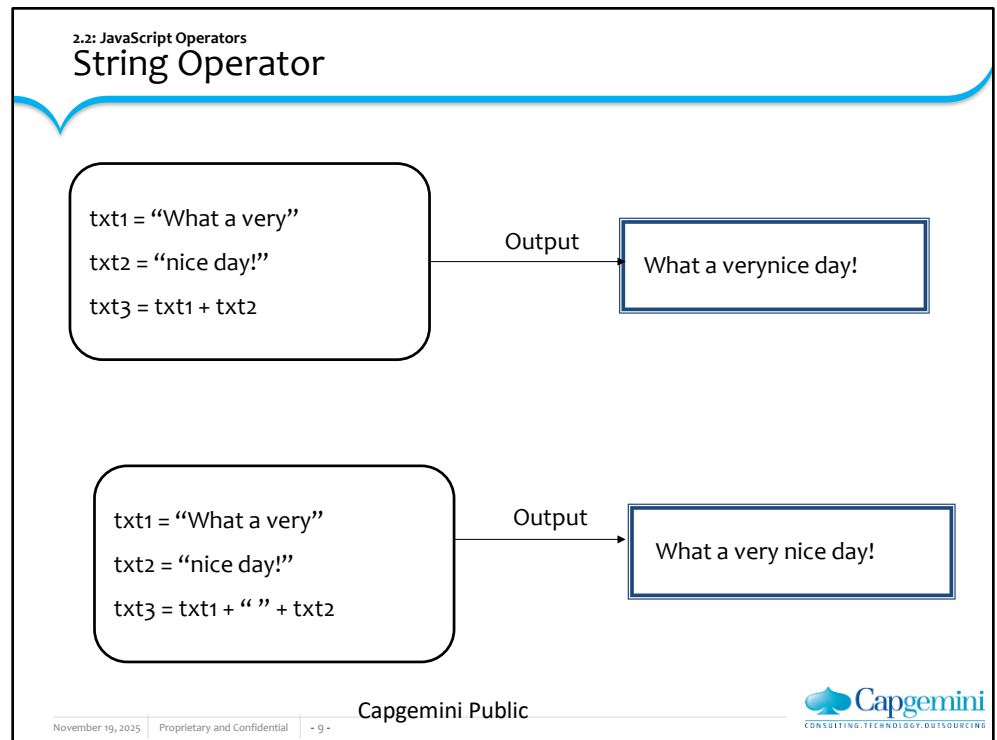
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2.2: JavaScript Operators

Logical Operator

Operator	Description	Example
&&	and	x = 6; y = 3 x < 10 && y > 1 returns true
	or	x = 6; y = 3 x < 10 y > 5 returns true
!	not	x = false !x returns true

Instructor Notes:**String Operator:**

```
txt1="What a very"
txt2="nice day!"
txt3=txt1+txt2
```

A string is most often a text, for example "Hello World!". To stick two or more string variables together, use the + operator.

The variable txt3 now contains "What a verynice day!". To add a space between two string variables, insert a space into the expression, OR in one of the strings.

```
txt1="What a very"
txt2="nice day!"
txt3=txt1+" "+txt2
```

Or

```
txt1="What a very "
txt2="nice day!"
txt3=txt1+txt2
```

The variable txt3 now contains "What a very nice day!".

Instructor Notes:

2.2: JavaScript Operators

Typeof Operator

typeof	undefinedvariable	“undefined”
typeof	33	“number”
typeof	“abcdef”	“string”
typeof	true	“boolean”
typeof	null	“object”

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Typeof Operator:

The typeof operator returns the type of data that its operand currently holds.

```
<HTML>
<HEAD>
<TITLE>Using typeof</TITLE>
</HEAD>
<BODY>
<SCRIPT LANGUAGE="JavaScript">
<!-- var num1=20
var str1="abc"
var bool1=true
var num2=null
var var1;
document.write("type of str1 : "+typeof(str1)+"<BR>")
document.write("type of num1 : "+typeof(num1)+"<BR>")
document.write("type of bool1 : "+typeof(bool1)+"<BR>")
document.write("type of num2 : "+typeof(num2)+"<BR>") -->
</SCRIPT>
</BODY>
</HTML>
```

Example 2.1 typeof operator (typeof_ex.html)

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Demo

➤ `Typeof_ex.html`



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Instructor Notes:

2.3: Control Structures and Loops

Control Structures and Loops

➤ JavaScript supports the usual control structures:

– the conditionals:

- if,
- if...else
- If ... else if ... else
- switch

– iterations:

- for
- while

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Control Structures and Loops

Conditional statements are used to perform different actions based on different conditions. Loops execute a block of code for specified number of times or while a specified condition is true.

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2.3: Control Structures and Loops
The if Statement

```
if(condition) {  
    statement 1  
} else {  
    statement 2  
}
```

```
if(a>10) {  
    document.write("Greater than 10")  
} else {  
    document.write("Less than 10")  
}
```

Shorthand

```
document.write( (a>10) ? "Greater than 10" : "Less than 10" );
```

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The if Statement:

The condition is any JavaScript expression that evaluates to the Boolean type, either true or false.

The example as shown on the slide.

A shorthand method can also be used for these types of statements, where ? indicates the 'if' portion and : indicates the 'else' portion. This statement is equivalent to the previous example:

The equivalent shorthand method is also seen on the slide.

Instructor Notes:

2.3: Control Structures and Loops

The Switch Statement

Syntax

```
switch (variable){  
  case outcome1:{  
    //stmts for outcome 1  
    break; }  
  case outcome2:{  
    //stmts outcome 2  
    break; }  
  ...  
  default: {  
    //none of the outcomes  
    is chosen }  
}
```

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2.3: Control Structures and Loops

The Switch Statement

➤ **Code Snippet**

```
switch (day) {  
  case "Monday": {  
    document.write("weekday")  
    break;}  
  case "Saturday": {  
    document.write("weekday")  
    break}  
  ...  
  default: {  
    document.write("Invalid day of the week")  
  }  
}
```

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2.3: Control Structures and Loops

The for Statement

> Syntax

```
for( [initial expression;][condition;][increment expression] )  
{  
    statements  
}
```

> Code Snippet

```
for(var i=0;i<10;i++){  
    document.write("Hello");}
```

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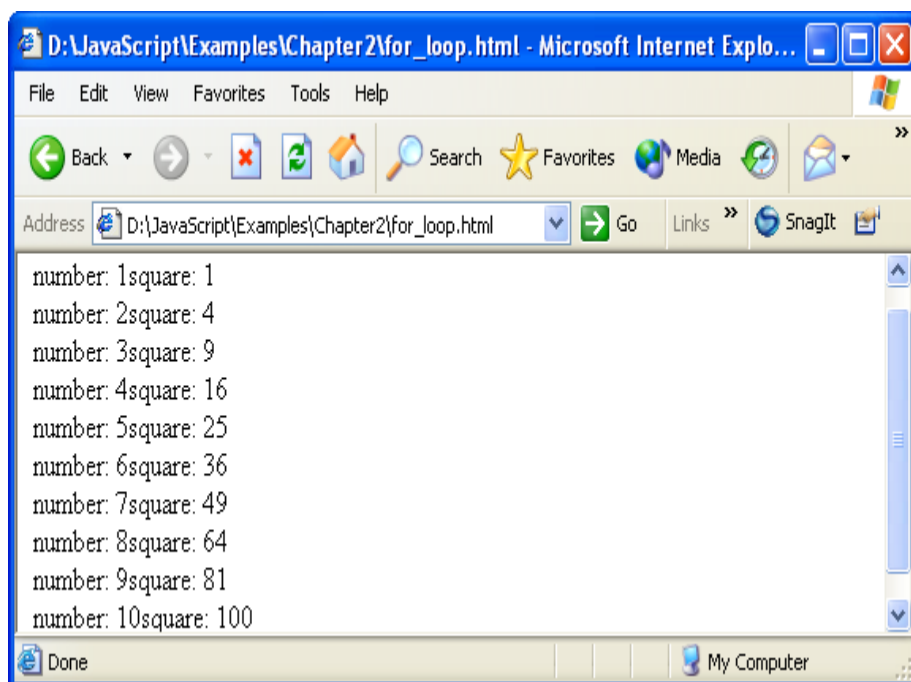
Looping Statements

The 'for' Statement:

```
<HTML>  
<HEAD>  
<SCRIPT LANGUAGE="JavaScript">  
<!-- hide script  
for (i=1; i<=10; i++)  
{  
    sq=i*i  
    document.write("number: " + i + "square: " + sq + "<BR>")  
}  
// end script hiding -->  
</SCRIPT>  
</HEAD>  
<BODY></BODY></HTML>
```

Example 2.2 For Construct (for_ex.html)

And it produces the output as:

Instructor Notes:**The 'while' Statement**

The while statement continues to repeat the loop as long as the condition is true. The syntax for the while statement is as follows:
while (condition):

```
{  
    statements  
}
```

```
<HTML>  
<HEAD>  
<SCRIPT LANGUAGE="JavaScript">  
<!-- hide script  
i=1  
while (i<=10)  
{  
    sq=i*i  
    document.write("number: " + i + "square: " + sq + "<BR>")  
    i++  
}  
// end script hiding -->  
</SCRIPT>  
</HEAD>  
<BODY></BODY>  
</HTML>
```

Example 2.3 While Construct
And it produces the output as in the previous screen shot.

Instructor Notes:

2.3: Control Structures and Loops

The break and continue Statements

➤ **break**

– Writing break inside a switch, for, while control structure will cause the program to jump to the end of the block. Control resumes after the block, as if the block had finished

➤ **continue**

– Writing continue inside a loop will cause the program to jump to the test condition of the structure and re-evaluate and perform instruction of the loop. Control resumes at the next iteration of the loop

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The break statement:

This statement is used to break out of the current ‘for’ or ‘while’ loop. Control resumes after the loop, as if it had finished.

The continue Statement:

This statement continues a ‘for’ or ‘while’ loop without executing the rest of the loop. Control resumes at the next iteration of the loop.

Instructor Notes:

Demo

➤ For_ex.html



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Instructor Notes:

2.4: JavaScript Functions

JavaScript Functions

➤

The function statement

```
function myFunction (arg1, arg2, arg3)
{
    statements
    return //The return keyword returns a value.
}
```

➤

How to call a function

```
myFunction( "abc", "xyz", 4 )
or
myFunction()
```

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The function Statement:

A function contains some code that will be executed by an event or a call to that function. A function is a set of statements. You can reuse functions within the same script, or in other documents. You define functions at the beginning of a file (in the head section), and call them later in the document.

The syntax of a typical function is as follows:

```
function myfunction(arg1, arg2, arg3)
{
    statements
}
```

How to Call a Function?

A function is not executed before it is called. You can call a function containing arguments:

```
myfunction("abc","xyz",4)
or without arguments:
myfunction()
```

Instructor Notes:

2.4: JavaScript Functions

Argument Arrays and How to call a Function

➤ **Syntax for the arguments array:**

```
arguments[index]  
functionName.arguments[index]
```

index – ordinal number of the argument starting at zero
arguments.length – Total number of arguments

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Using the arguments Array

The arguments of a function are maintained in an array. Within a function, you can address the parameters passed to it as follows:

`arguments[i]`

`functionName.arguments[i]`

where *i* is the ordinal number of the argument, starting at zero. So, the first argument passed to a function would be `arguments[0]`. The total number of arguments is indicated by `arguments.length`. Using the arguments array, you can call a function with more arguments than it is formally declared to accept. This is often useful if you don't know in advance how many arguments will be passed to the function. You can use `arguments.length` to determine the number of arguments actually passed to the function, and then treat each argument using the arguments array.

Instructor Notes:

The Function Statement (Contd..)

➤ Syntax

```
function myConcat(separator) {  
    result = ""  
    for(var index=1; index<arguments.length;index++){  
        result += arguments[index] + separator  
    }  
    return result  
}
```

➤ Code Snippet

```
myConcat( " , " , "red" , "orange" , "blue" )  
// returns "red, orange, blue"
```

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For example, consider a function that concatenates several strings. The only formal argument for the function is a string that specifies the characters that separate the items to concatenate. The function is defined as shown on the slide. You can pass any number of arguments to this function, and it creates a list using each argument as an item in the list.

```
// returns "red, orange, blue, "  
myConcat( " , " , "red" , "orange" , "blue" )  
// returns "elephant; giraffe; lion; cheetah;"  
myConcat( " ; " , "elephant" , "giraffe" , "lion" , "cheetah" )  
// returns "sage. basil. oregano. pepper. parsley. "  
myConcat( " . " , "sage" , "basil" , "oregano" , "pepper" , "parsley" )
```

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions

➤ **eval:**

- Evaluates a string of JavaScript code without reference to a particular object.

```
eval(expr)
where expr is a string to be evaluated
```

➤ **isFinite:**

- Evaluates an argument to determine whether it is a finite number.

```
isFinite(number)
where number is the number to evaluate
```

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Predefined Functions:

JavaScript has several top-level predefined functions:

Eval, isFinite, isNaN, parseInt and parseFloat, Number and String

eval Function

The eval function evaluates a string of JavaScript code without reference to a particular object. The syntax of eval is:

eval(expr) where expr is a string to be evaluated.

If the string represents an expression, eval evaluates the expression. If the argument represents one or more JavaScript statements, eval performs the statements. Do not call eval to evaluate an arithmetic expression; JavaScript evaluates arithmetic expressions automatically.

isFinite Function

The isFinite function evaluates an argument to determine whether it is a finite number. The syntax of isFinite is:

isFinite(number) where number is the number to evaluate.

If the argument is NaN, positive infinity or negative infinity, this method returns false, otherwise it returns true. The following code checks client input to determine whether it is a finite number.

```
if(isFinite(ClientInput) == true)
{
    /* take specific steps */
}
```

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions (Contd..)

>

isNaN :

- Evaluates an argument to determine if it is “NaN” (not a number)

isNaN (testValue)
where testValue is the value you want to evaluate

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> 24 <

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isNaN Functions:

The isNaN function evaluates an argument to determine if it is "NaN" (not a number). The syntax of isNaN is:

isNaN(testValue)

where testValue is the value you want to evaluate.

The parseFloat and parseInt functions return "NaN" when they evaluate a value that is not a number. isNaN returns true if passed "NaN," and false otherwise.

The following code evaluates floatValue to determine if it is a number and then calls a procedure accordingly:

```
floatValue=parseFloat(toFloat)
if (isNaN(floatValue)) {
    notFloat()
} else {
    isFloat()
}
```


Instructor Notes:

2.4: JavaScript Functions

Predefined Functions (Contd..)

➤

parseInt and parseFloat

- Returns a numeric value for string argument.

parseInt(str)
parseFloat(str)

parseInt(str, radix)
//returns an integer of specified radix of the string argument

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parseInt and parseFloat Functions:

The two "parse" functions, `parseInt` and `parseFloat`, return a numeric value when given a string as an argument.

The syntax of `parseFloat` is:

`parseFloat(str)`

where `parseFloat` parses its argument, the string `str`, and attempts to return a floating-point number. If it encounters a character other than a sign (+ or -), a numeral (0-9), a decimal point, or an exponent, then it returns the value up to that point and ignores that character and all succeeding characters. If the first character cannot be converted to a number, it returns "NaN" (not a number).

The syntax of `parseInt` is:

`parseInt(str [, radix])`

where `parseInt` parses its first argument, the string `str`, and attempts to return an integer of the specified radix (base), indicated by the second, optional argument, `radix`. For example, a radix of ten indicates to convert to a decimal number, eight octal, sixteen hexadecimal, and so on. For radices above ten, the letters of the alphabet indicate numerals greater than nine. For example, for hexadecimal numbers (base 16), A through F are used.

If `parseInt` encounters a character that is not a numeral in the specified radix, it ignores it and all succeeding characters and returns the integer value parsed up to that point. If the first character cannot be converted to a number in the specified radix, it returns "NaN." The `parseInt` function truncates the string to integer values.

Page 02-25

Instructor Notes:

2.4: JavaScript Functions

Predefined Functions (Contd..)

>

Number and string

- Converts an object to a number or a string.

Number(objectReference)

String(objectReference)

today = new Date(430054663215)

now = String(today)

// returns "Thu Aug 18 04:37:43 GMT-0700 (PDT) 1983"

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Number and String Functions:

The Number and String functions let you convert an object to a number or a string.

The syntax of these functions is:

Number(objRef)

String(objRef)

where objRef is an object reference. The following example converts the Date object to a readable string.

```
D = new Date(430054663215)
```

```
// The following returns
```

```
// "Thu Aug 18 04:37:43 GMT-0700 (Pacific Daylight Time) 1983"
```

```
x = String(D)
```

Instructor Notes:

2.4: JavaScript Functions

Global and Local Variables

➤ Code Snippet for scope of variables

```
<script>
  var companyName="CAPGEMINI"
  function displayName(){
    var employeeName="Tom"
    document.write("Welcome to "+companyName+", "+employeeName)
  }
</script>
```

Global Variable

Local Variable

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**Scope of Variables:**

JavaScript supports two variable scopes:

Global variables

Local variables

The local variable applies only within a function and limits the scope of the variable to that function. To declare a local variable, the variable name must be preceded by var, as shown following:

```
var MaxValue=0;
```

Any variable declaration that is not within a function, is treated as a global variable. The syntax to declare a global variable is the same as that for local variable.

Instructor Notes:

2.4: JavaScript Functions

Global and Local Variables

- Variables that exist only inside a function are called **Local variables**
- The values of such **Local variables** cannot be changed by the main code or other functions
- Variables that exist throughout the script are called **Global variables**
- Their values can be changed anytime in the code and even by other functions

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**Using Global and Local Variables:**

You can choose between local and Global variables by using the following guidelines :

If the value of a variable is meant to be used by any part of the program, both inside and outside functions, the variable should be declared outside any function. This has the effect of making it global and modifiable by any part of the program. The best place to declare global variables is in the <head> block of the HTML document.

If the variable is needed only within a particular function, the variable should be declared inside that function.

If you want the value of a variable to be modified only by the main script of a single function, but you need to use it in another function, pass the variable as an argument to that function. This has the effect of making a copy of the variable and assigning its value to the argument. As the function works and modifies its own copy of the variable, it will not effect the original. Argument variables are automatically declared as local to that function. Even if the argument has the same name as the variable being passed, making changes to it does not effect the variable that was passed. The only exception to this is objects. When an object is passed as an argument, it is passed by reference as opposed to being passed by value. Instead of making a copy of the object, the function uses the original object. Changes made to an object's properties within the function have an effect on the original object.

Instructor Notes:

Demo

- If_else.html
- Switch_ex.html
- For_ex.html
- Break_con_ex.html
- Fun_ex.html
- Num_string_fun.html



Instructor Notes:

Lab

- Lab 2 :
- The JavaScript language



Instructor Notes:

Summary

- **Data Types & Variables**
 - Numbers, Strings, Boolean, and Null
- **Operators & Expressions**
- **Functions**
- **Predefined Functions**
 - eval, isFinite, isNaN, parseInt & parseFloat, Number & String
- **Global and Local variables**



Instructor Notes:**Answers**

1. Option o
2. True

Review Question

- **Question 1: Which of the following two variable scopes is supported by JavaScript:**

- Global, Local
- Functional, Non functional
- Static, Dynamic



- **Question 2: The eval function evaluates a string of JavaScript code without reference to a particular object.**

- True/False

Instructor Notes:**Answers**

1. 4
2. 5
3. 1
4. 2
5. 3

Review Question: Match the Following

1. Loop statements
2. Arithmetic operators
3. Predefined function
4. Assignment operators
5. Logical operators

1. isNaN
2. +=, -=
3. &&,
4. For, While
5. ++, --, %, *

